Hotel Booking Analytics

Capstone Project – Business Analytics System

Reducing cancellation losses and forecasting demand for a City Hotel and a Resort Hotel in Portugal

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Contents

1. Executive Summary

This capstone builds a **business analytics system** for two Portuguese hotels – a City Hotel in Lisbon and a Resort Hotel in the Algarve – using 119,208 cleaned bookings with arrival dates from July 2015 to August 2017. The system combines a cleaned star-schema dataset, a six-page interactive **Excel dashboard** (the main tool), a matching Power BI report design, a rule-based cancellation risk flag and a 12-month demand forecast. It follows the Business Analytics approach: the focus is on explaining past performance and turning it into clear actions, with a simple, explainable forecast built in Excel.

Headline metric	Value
Bookings analysed	119,208
Cancellation rate	37.1% (44,198 bookings)
Realised room revenue	€26.0M
Booked value lost to cancellations	€16.7M
Average daily rate (stayed bookings)	€100.17
Average length of stay	3.39 nights
Repeat guests	3.1%

Table 1. Key performance indicators (all hotels, Jul 2015 – Aug 2017).

Main findings

- Cancellations are the biggest business problem: 37.1% of bookings cancel, and the lost booked value (€16.7M) equals 64% of the revenue actually earned. The City Hotel cancels far more (41.8%) than the Resort Hotel (27.8%).
- Cancellation risk rises steadily with lead time: 9.6% for bookings made within a week, 67.7% for bookings made more than a year ahead.
- Group bookings (61.1%) and online travel agents (36.8%) cancel most; direct bookings cancel least (15.4%).
- Guests who make special requests or ask for parking are far more committed – 47.8% cancel with no requests versus 22.0% with one request.
- The Resort Hotel is highly seasonal: its average daily rate rises from about €49 in January to about €181 in August.
- A simple rule-based risk flag (long lead time + Groups/Online TA + no special requests) separates risky bookings well: High-flag bookings actually cancelled 71.9% of the time, Low-flag bookings only 18.6%.
- The forecast expects next-year (Sep 2017 – Aug 2018) revenue of €8.9M for the City Hotel (+11.0%) and €5.9M for the Resort Hotel (+8.5%).

Top recommendations: (1) require a deposit or pre-payment for bookings with a High risk flag; (2) use the number of High and Medium flagged bookings per arrival date to set safe overbooking levels; (3) grow direct bookings and tighten group and OTA cancellation terms; (4) engage long-lead guests with reminders and pre-arrival requests; (5) apply seasonal pricing, especially at the Resort Hotel. Preventing just 20% of cancellations would recover about €3.3M of booked value.

2. Business Context and Data

The dataset is the Hotel Booking Demand dataset published by Antonio, de Almeida and Nunes (2019) in Data in Brief. It holds real booking records from two hotels in Portugal: a resort hotel in the Algarve and a city hotel in Lisbon. For this project it was supplied as five related tables (one fact table and four lookup tables) plus a data dictionary. Currency is not stated in the dictionary; because both hotels are in Portugal, values are treated as euros (€).

Table	Rows	Role	Key fields
Bookings	119,390	Fact – one row per booking	lead_time, arrival date, nights, guests, adr, is_canceled, country, special requests
Hotels	2	Lookup	Hotel_ID → City Hotel / Resort Hotel
Market Segments	26	Lookup	market_segment + distribution_channel
Customer Types	36	Lookup	customer_type + deposit_type + meal
Reservation Status	3	Lookup	Check-Out / Canceled / No-Show

Table 2. Source tables.

Revenue management teams at hotels care about three things this data can answer: how many rooms will be filled, at what price, and how many bookings will disappear before arrival. Cancellations are especially costly because a room held for a guest who cancels late often cannot be resold.

3. Problem Statement

> The two hotels lose more than one in three bookings to cancellation (37.1%). Between July 2015 and August 2017 this represented €16.7M of booked room value that was never earned – equal to 64% of the €26.0M actually realised. Management has no single view showing where cancellations come from, which upcoming bookings are at risk, and how much demand to expect, so pricing, deposit and overbooking decisions are made reactively. **This project builds a business analytics system (Excel + Power BI), supported by a simple cancellation risk flag and a demand forecast, to help revenue managers reduce cancellation losses, price by season and segment, and plan capacity with confidence.**

4. Research Objectives

Five research objectives guide the analysis. Four are descriptive (what happened and why) and one looks ahead (what will happen next). Each has its own page in the Excel dashboard (and the matching Power BI report).

#	Research objective	Answered in
RO1	Measure overall booking performance (volume, revenue, ADR, length of stay) and seasonal patterns for each hotel over time.	Sections 7.1 and 7.3; Power BI pages 1 and 3
RO2	Identify the main drivers of cancellations and quantify the revenue lost.	Section 7.2 and 8.2; Power BI page 2
RO3	Compare market segments and distribution channels on volume, value and reliability.	Section 7.4; Power BI pages 1–3
RO4	Profile guests (country, repeat status, special requests, stay length) to find loyalty and engagement opportunities.	Section 7.5; Power BI page 4
RO5	Forecast monthly bookings and revenue for the next 12 months for each hotel.	Section 8.1; Power BI page 5

Table 3. Research objectives.

5. Methodology

The project follows the CRISP-DM process: business understanding → data understanding → data preparation → analysis → evaluation → deployment. Tools were chosen so that the final system can be used by non-technical hotel staff.

Stage	What was done	Tool
Data preparation	Joined 5 tables, cleaned, engineered 17 new fields, built star schema	Python (pandas); Power Query steps documented
Descriptive analysis	KPIs and breakdowns by hotel, time, segment, channel, deposit, lead time, guest profile	Excel (COUNTIFS/SUMIFS), Power BI (DAX)
Forecast & risk flag	12-month forecast (same month last year × damped growth); rule-based cancellation risk flag from the analysis findings	Excel formulas; Power BI DAX columns
Reporting system	Main: 6-page Excel dashboard with 4 shared drop-down filters, charts, heatmaps and what-if. Second: matching 6-page Power BI report	Excel (main); Power BI Desktop
Deployment plan	Scheduled refresh, quarterly review of the risk rule, row-level security	Power BI Service

Table 4. Methodology and tools.

System architecture

Raw CSV tables → cleaning & feature engineering → **star schema** (Fact_Bookings, Fact_Forecast, 6 dimension tables) → two front-ends: the **Excel dashboard** (main tool – formula-driven pages, forecast and risk flag) and the **Power BI report** (interactive pages, drill-through, what-if). A rule-based risk flag (Low / Medium / High) is calculated for every booking in both front-ends, using formulas that managers can read and change.

6. Data Preprocessing

All cleaning steps were logged (Excel sheet "Cleaning_Log"). Only 182 rows (0.15%) were removed, so the analysis keeps almost all real business activity.

Step	Action	Detail	Rows
1	Load	Bookings table loaded (raw rows)	119,390
2	Check keys	All foreign keys match lookup tables (Hotel, Segment, Customer Type, Reservation)	0
3	Missing values	children blank -> 0	4
4	Missing values	country blank -> "UNK" (Unknown)	488
5	Missing values	agent blank -> 0 = "No agent" (booked without travel agent)	16,340
6	Missing values	company blank -> 0 = "No company" (not a corporate booking)	112,593
7	Remove invalid	Bookings with 0 guests (adults+children+babies=0)	180
8	Remove invalid	Negative ADR	1
9	Remove outlier	ADR above 1,000 (single 5,400 entry error)	1
10	Check duplicates	Identical rows (excluding Booking_ID) kept: there is no guest ID, so they are most likely separate rooms in the same group booking.	32,995
11	Check consistency	is_canceled matches reservation status (1 = Canceled/No-Show)	0
12	Feature engineering	Added arrival_date, booking_date, total_nights, total_guests, has_kids, booking_value, realised_revenue, lost_revenue, lead_time_band, stay_length_band, season, weekday, has_agent, has_company flags	119,208
13	Final	Clean fact table rows	119,208
14	Final	Rows removed in total	182

Table 5. Data cleaning log.

Key decisions and reasons

- **Missing agent / company** (14% / 94%) are not errors – they mean the booking was not made through an agent or a company. They were set to 0 and turned into yes/no flags.
- **Missing country** (488 rows) was kept as "Unknown" instead of deleting bookings.
- **Zero-guest bookings** (180) and **impossible prices** (one negative ADR, one ADR of 5,400) were removed as data-entry errors.
- **Duplicate-looking rows** (32,995) were kept for business reporting because there is no guest ID – identical rows are usually separate rooms in the same group booking. Removing them would under-count real business activity.
- **Consistency check:** is_canceled matches the reservation status for every row (1 = Canceled or No-Show).

Feature engineering

New field	Formula / rule	Why
arrival_date	Year + month + day combined	Time-series analysis, link to calendar
booking_date	arrival_date − lead_time	Analyse when bookings are made
total_nights / total_guests	weekend + week nights; adults + children + babies	Stay length, party size
booking_value	adr × total_nights	Value of each booking
realised_revenue / lost_revenue	booking_value split by is_canceled	Revenue actually earned vs lost
lead_time_band, stay_length_band, season	Grouped ranges	Easy-to-read segments for managers
has_agent, has_company,	Yes/no flags	Filters and segments

New field	Formula / rule	Why
has_kids, has_previous_cancel		
risk_factors, risk_flag	1 point each: lead time > 90 days, Groups/Online TA, no special requests; 0–1 Low, 2 Medium, 3 High	Simple, explainable cancellation risk flag

Table 6. Engineered fields.

Data model (star schema)

Fact_Bookings (119,208 rows) connects to Dim_Hotel, Dim_Market_Segment, Dim_Customer_Type, Dim_Reservation_Status, Dim_Country and Dim_Date (active on arrival date, inactive on booking date). Fact_Forecast connects to Dim_Hotel and Dim_Date. This design keeps Power BI fast and makes every measure filterable by every dimension.

7. Descriptive Analysis and Findings

7.1 Overall performance (RO1)

Hotel	Bookings	Cancel rate	Realised revenue	Lost revenue	ADR (stayed)	Avg stay
City Hotel	79,162.0	41.8%	€14.4M	€10.9M	€106.04	2.92
Resort Hotel	40,046.0	27.8%	€11.6M	€5.8M	€90.83	4.14

Table 7. Performance by hotel.

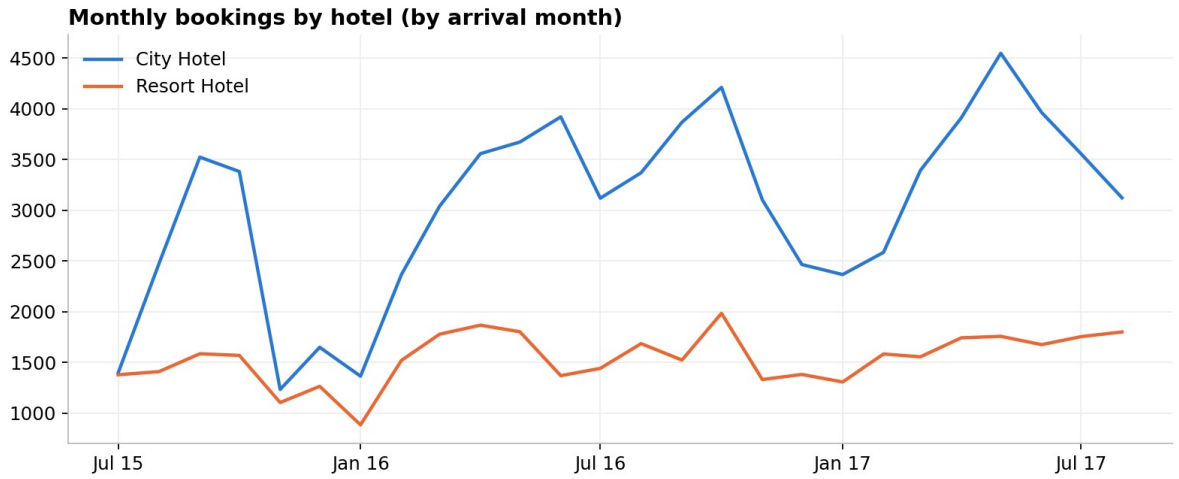


Figure 1. Monthly bookings by hotel.

The City Hotel handles two-thirds of bookings and 55% of realised revenue. Both hotels grew from 2015 to 2017; the City Hotel shows a clear spring–autumn peak, while the Resort Hotel is flatter in volume but earns much more per night in summer.

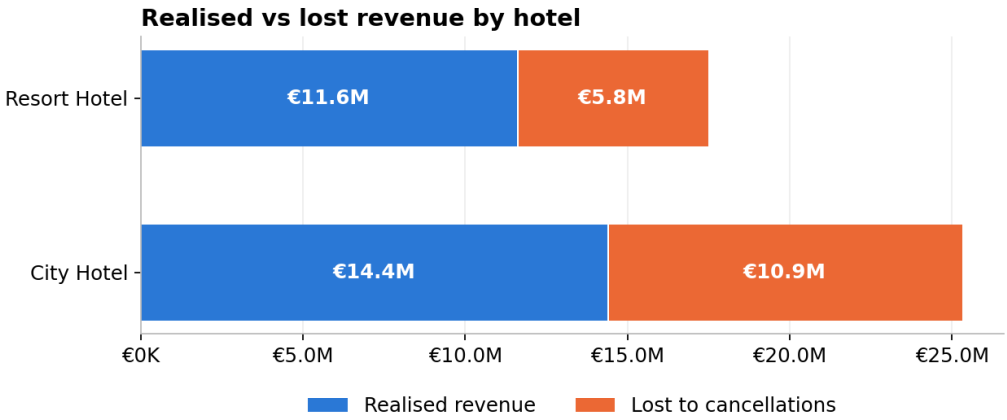


Figure 2. Realised vs lost revenue by hotel.

7.2 Cancellation drivers (RO2)

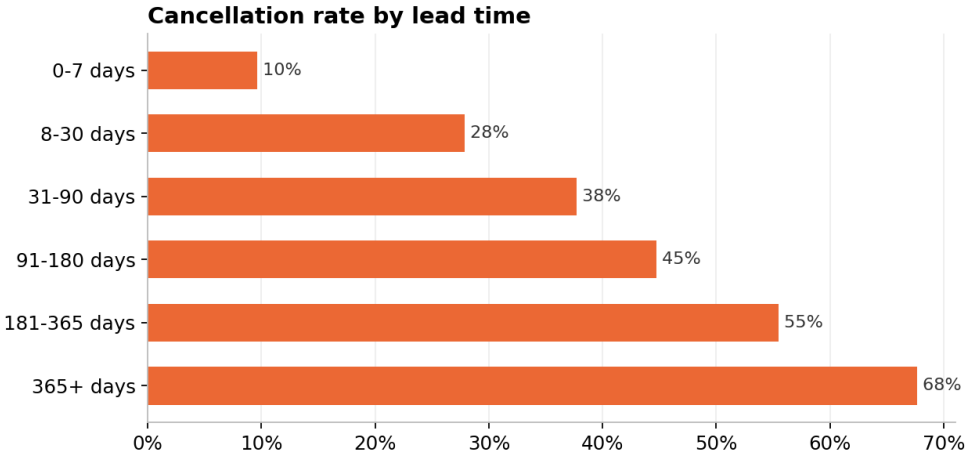


Figure 3. Cancellation rate rises with lead time.

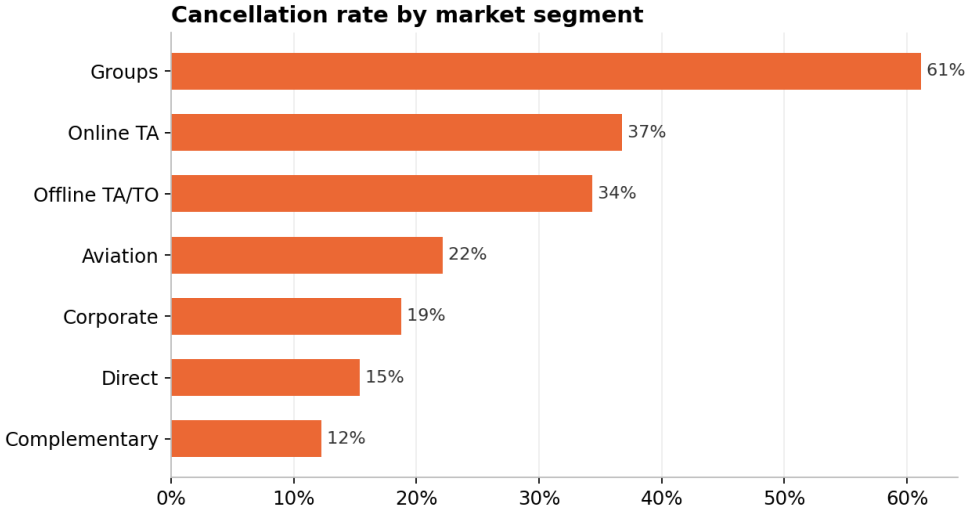


Figure 4. Cancellation rate by market segment.

Driver	Lowest risk	Highest risk
Lead time	0–7 days: 9.6%	365+ days: 67.7%
Market segment	Direct: 15.4%	Groups: 61.1%
Deposit type	Refundable: 22.2%	Non Refund: 99.4%
Special requests	3+ requests: ≤17.8%	0 requests: 47.8%
Car parking requested	1+ space: 0.0%	No parking: 39.5%
Guest history	Repeat guest: 14.7%	Had a previous cancellation: 91.7%
Country	Germany: 16.7%	Portugal: 56.7%

Table 8. Cancellation rate by driver.

Data quirk – "Non Refund" bookings cancel 99.4% of the time. This is the opposite of what a non-refundable rate should do. 97% of these bookings come from Portugal and most are Groups or Offline travel agents, which suggests blocks of rooms held by agents under this code and later released. It should be checked with the hotels before using deposit type in pricing decisions.

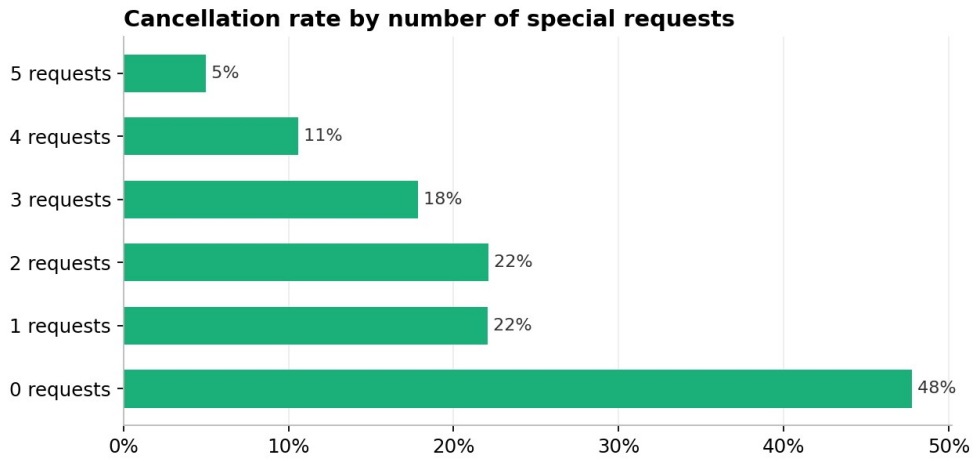


Figure 5. More special requests = fewer cancellations.

7.3 Seasonality and pricing (RO1)

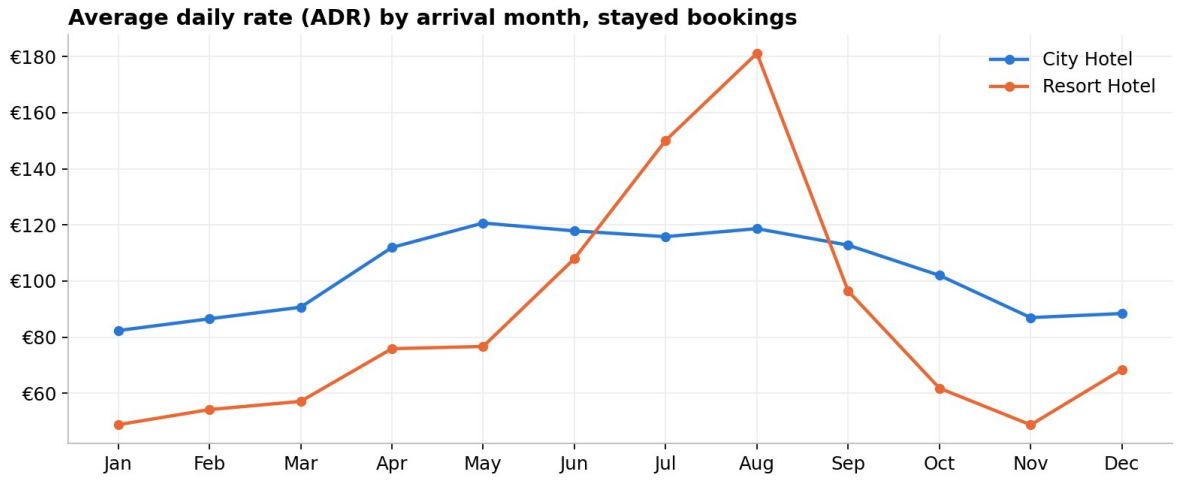


Figure 6. ADR by arrival month.

Season	Bookings	Cancel rate	ADR (stayed)	Realised revenue
Spring (Mar-May)	32,624.0	37.8%	€93.77	€6.1M
Summer (Jun-Aug)	37,434.0	38.8%	€130.16	€11.6M
Autumn (Sep-Nov)	28,418.0	36.9%	€90.03	€5.2M
Winter (Dec-Feb)	20,732.0	33.1%	€73.06	€3.0M

Table 9. Performance by season (both hotels).

Summer (Jun–Aug) delivers 45% of all realised revenue with the highest ADR (€130). The Resort Hotel's rate is almost four times higher in August than in January, while the City Hotel's rate moves only between about €82 and €121. Winter is the cheapest and least cancelled season – a good target for packages and promotions.

7.4 Market segments and channels (RO3)

Segment	Bookings	Cancel rate	Realised revenue	Lost revenue	ADR	Avg lead (days)
Online TA	56,408	36.8%	€13.7M	€10.2M	€114	83
Offline TA/TO	24,181	34.3%	€5.7M	€2.5M	€84	135
Groups	19,790	61.1%	€1.9M	€2.8M	€77	187
Direct	12,582	15.4%	€4.1M	€993K	€114	50

Segment	Bookings	Cancel rate	Realised revenue	Lost revenue	ADR	Avg lead (days)
Corporate	5,282	18.8%	€578K	€196K	€67	22
Complementary	728	12.2%	€5K	€0K	€3	14
Aviation	235	22.1%	€71K	€17K	€101	4

Table 10. Market segment performance.

Online travel agents (OTA) are the largest segment by volume and revenue, with a high ADR (€114) but also €10.2M of lost value. Direct bookings have the same ADR as OTA with less than half the cancellation rate and no commission – the most valuable channel to grow. Groups book earliest (187 days ahead) and cancel most, costing more in lost value (€2.8M) than they earn (€1.9M).

7.5 Guest profile (RO4)

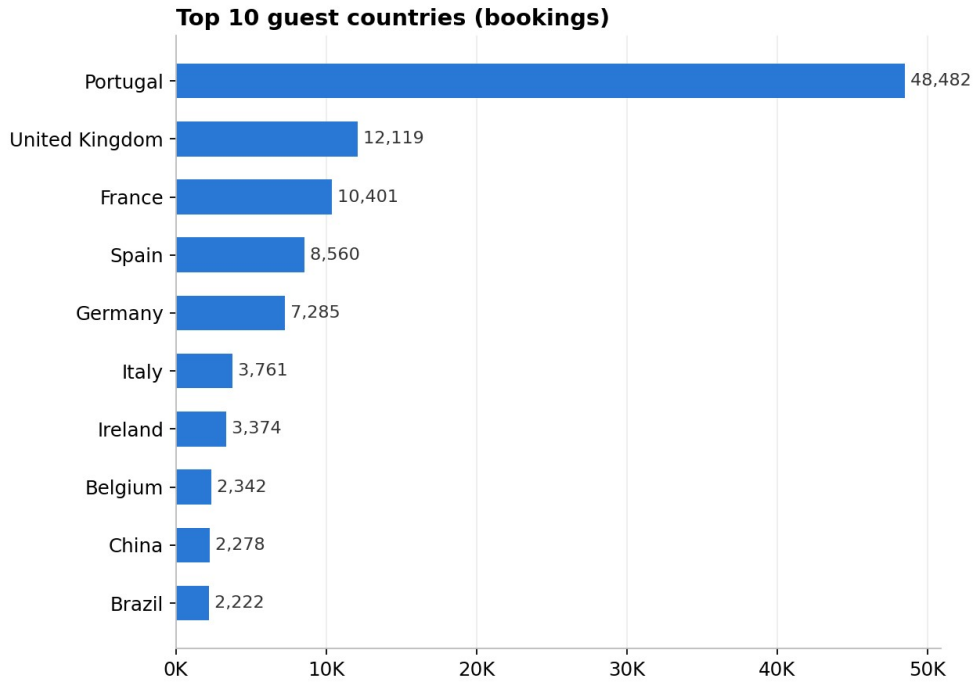


Figure 7. Top 10 guest countries.

- Guests come from 178 countries; Portugal (41%), the UK, France, Spain and Germany make up 73% of bookings.
- Portuguese bookings cancel 56.7% of the time versus 23.6% for all other countries combined.
- Only 3.1% of bookings are from repeat guests, who cancel much less (14.7%) – there is room for a loyalty programme.
- Most stays are short (1–3 nights = 64% of bookings); stays of 15+ nights are rare but cancel 56% of the time.

8. Forecast and Risk Flag

8.1 Demand and revenue forecast (RO5)

Method: Forecast for a month = actual value in the same month last year × growth factor, where growth factor = 1 + 50% × (last 6 months ÷ same 6 months a year earlier − 1). This keeps each hotel's seasonal shape and adds half of the recent growth trend ("damped growth") so one strong half-year is not over-projected. It is simple enough to run in Excel with ordinary formulas (sheet "Forecast") and to explain to managers.

Backtest: the method was tested by pretending it was February 2017 and forecasting March–August 2017, then comparing with what happened. Mean absolute percentage error (MAPE):

Hotel	Metric	MAPE – damped growth	MAPE – same as last year
City Hotel	Bookings	11.4%	10.0%
City Hotel	Revenue	8.4%	17.9%
Resort Hotel	Bookings	12.1%	11.1%
Resort Hotel	Revenue	6.1%	15.5%

Table 11. Forecast backtest accuracy (lower is better).

The damped-growth method is much better for revenue (6–8% error vs 15–18%). For booking counts both methods are similar (10–12%), because booking volume grew less than revenue – most growth came from higher prices.

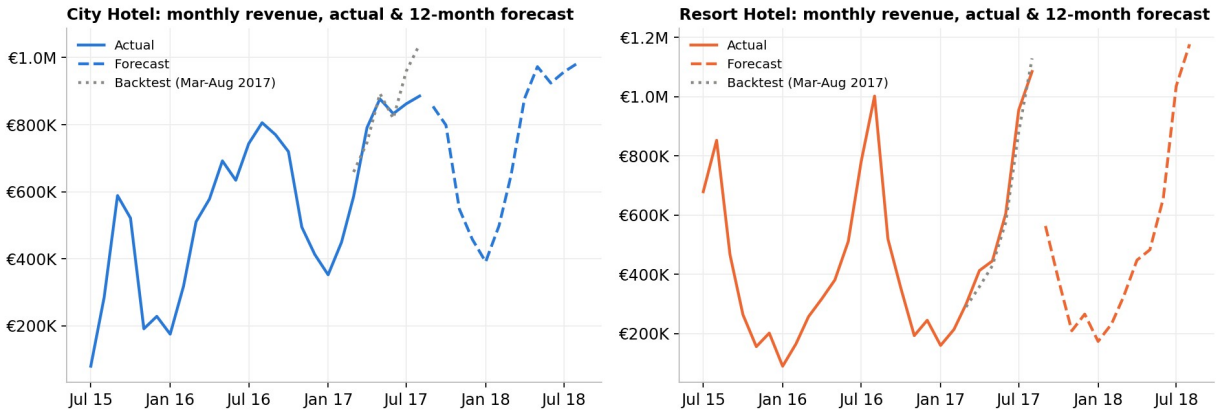


Figure 8. Monthly revenue: actual and 12-month forecast.

Hotel	Metric	Last 12 months (Sep 16–Aug 17)	Forecast (Sep 17–Aug 18)	Change
City Hotel	Bookings	41,094	42,896	4.4%
City Hotel	Realised revenue	€8.0M	€8.9M	11.0%
Resort Hotel	Bookings	19,392	19,725	1.7%
Resort Hotel	Realised revenue	€5.5M	€5.9M	8.5%

Table 12. 12-month forecast summary.

8.2 Rule-based cancellation risk flag (supports RO2)

To turn the cancellation findings into action, every booking gets a simple risk flag. It uses the three strongest and most reliable drivers from Section 7.2 – one point for each:

Risk factor	Rule	Evidence from the analysis
Long lead time	Lead time over 90 days	Cancel rate rises from 9.6% (0–7 days) to 67.7% (365+ days)
High-cancel segment	Groups or Online TA	Groups 61.1%, Online TA 36.8% vs Direct 15.4%
No special requests	0 requests	47.8% cancel vs 22.0% with one request

Risk factor	Rule	Evidence from the analysis
Flag	0–1 points = Low · 2 = Medium · 3 = High	

Table 13. Risk flag rules.

Deposit type and parking were deliberately **not** used because of the possible data-recording issues described in Section 7.2. The rule is built in Excel (live formulas with an editable lead-time threshold) and in Power BI (calculated columns), so managers can see exactly why a booking is flagged.

Risk flag	Bookings	Share	Actual cancel rate	Lost value
Low	54,299	45.5%	18.6%	€3.8M
Medium	44,482	37.3%	43.6%	€7.7M
High	20,427	17.1%	71.9%	€5.1M

Table 14. Results of the risk flag on all bookings.

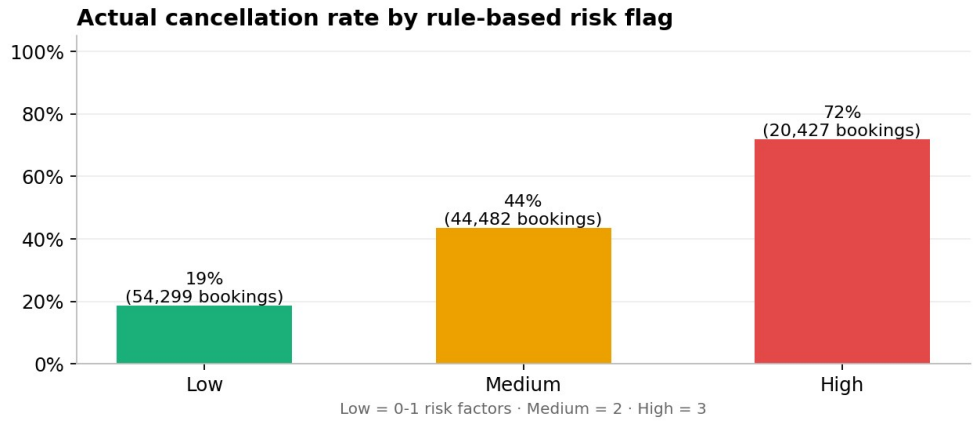


Figure 9. Actual cancellation rate by risk flag.

The flag works: High-flag bookings actually cancelled 71.9% of the time – almost four times the 18.6% of Low-flag bookings. The High group (20,427 bookings) lost €5.1M in booked value while earning only €2.0M. At the City Hotel the High flag reaches 77.7% actual cancellations.

Why a rule instead of a machine-learning model? This is a Business Analytics project. A rule based on three clear findings is easy for managers to understand, check and change, needs no special software, and updates automatically in Excel and Power BI. A machine-learning model could be added later (Section 12).

9. The Business Analytics System

9.1 Excel dashboard – the main tool (Hotel_Booking_Analytics.xlsx)

Excel was chosen as the main tool because every manager already has it, every number is a visible formula, and inputs such as the what-if %, the risk threshold and the forecast damping are easy to change. The workbook works like a Power BI report: six dashboard pages, navigation buttons, and four drop-down filters (Hotel, Year, Market segment, Risk flag) on the Overview page that update every page at once.

Sheet	Purpose
Overview	KPI cards, monthly trend charts, hotel comparison, realised vs lost revenue by segment, key-insight box. Holds the 4 shared filters.
Cancellations	Cancellation rate by lead time, deposit, segment, special requests and customer type (data bars + charts), segment × hotel heatmap, what-if calculator.
Revenue	ADR and revenue by month and hotel (heatmaps + line/column charts), season, customer type, meal plan.
Guests	Top 10 countries, new vs repeat guests, stay length, channels, families.
Forecast_Risk	Risk flag results and rules, 12-month forecast summary and actual-vs-forecast chart.
Explorer	Space for PivotTables, slicers, timeline and PivotCharts; double-click to drill down to bookings.
Forecast / Risk_Flag	Calculation sheets: damped-growth forecast with backtest; editable risk rule.
Cleaning_Log / Data	Audit trail of cleaning; 119,208-row Excel Table with named ranges used by every formula.

Table 15. Excel dashboard contents.

How it works: each Data column is a named range (e.g. d_Hotel). The four filters feed helper cells (f_hotel, f_year, f_seg, f_risk) that turn "All" into the wildcard "*". Every KPI, table and chart is a COUNTIFS / SUMIFS / AVERAGEIFS formula using these four filters, so one change updates the whole dashboard – the same idea as synced slicers in Power BI. Full instructions are in the **Excel Dashboard Guide**.

9.2 Power BI report – second tool and scale-up path

The Power BI report has six pages (Executive Overview, Cancellation Analysis, Revenue & Pricing, Guest Insights, Forecast & Risk Flag, Booking Detail drill-through). It uses the star schema, 20+ DAX measures (including time intelligence and a what-if parameter for "cancellation reduction"), three calculated columns for the risk flag, synced slicers, report-page tooltips, a Key Influencers AI visual, a decomposition tree and a Q&A box. Full build steps, relationships and every DAX measure are in the separate **Power BI Build Guide**, and an interactive web preview shows how the finished report looks.

9.3 How the system supports decisions

Decision	Who	Where in the system
Which upcoming bookings need a deposit or reminder?	Reservations team	Forecast_Risk page; Data filtered on Risk_Flag = High
How many rooms can we safely overbook next month?	Revenue manager	Count of High/Medium-flag bookings by arrival date
Which channel should get marketing budget?	Sales & marketing	Segment table, realised vs lost revenue
What price should we set for August vs January?	Revenue manager	ADR by month, seasonal revenue
How much could a new cancellation policy earn?	General manager	What-if cell on the Cancellations page
How many staff do we need next season?	Operations	12-month bookings forecast

Table 16. Decisions supported.

10. Recommendations

Recommendation	Action	Evidence / expected impact
1. Deposit for high-risk bookings	Ask for a deposit or pre-payment when a booking has a High risk flag (lead time over 90 days, Groups/Online TA, no special requests).	High-flag bookings cancelled 72% and lost €5.1M. Preventing 20% of these = about €1.0M.
2. Smart overbooking	Count High- and Medium-flag bookings for each arrival date and apply their typical cancel rates (72% / 44%) to set a safe overbooking level instead of a fixed %.	Fills rooms that would otherwise go empty after late cancellations.
3. Grow direct bookings	Offer best-rate guarantee, member discount or free breakfast on the hotel website.	Direct: same ADR as OTA (€114), 15% vs 37% cancellation, no commission.
4. Tighter group & OTA terms	Staged group deposits and room-block release dates; shorter free-cancellation windows with OTAs.	Groups lost €2.8M vs earned €1.9M.
5. Engage long-lead guests	Reminder emails at 60/30/7 days; invite guests to add requests, parking or dining.	Guests with 1+ request cancel 22% vs 48% with none.
6. Seasonal pricing & winter packages	Dynamic pricing at the Resort (Jan ≈ €49 → Aug ≈ €181); winter packages for both hotels.	Summer = 45% of revenue; winter has lowest cancellations.
7. Loyalty programme	Reward repeat stays; target UK, France, Germany guests who cancel less.	Repeat guests cancel 14.7% but are only 3% of bookings.
8. Fix the data	Check the "Non Refund" code and when parking requests are recorded.	Ensures decisions and future rules rest on correct data.

Table 17. Actionable recommendations.

11. Alignment with the Assessment Rubric

Marker (weight)	How this project meets it
Relevance & alignment with business goals (30%)	Clear problem statement on the €16.7M cancellation loss; 5 research objectives each mapped to a dashboard page; decisions table (Table 16).
Data visualization & reporting (25%)	Six Excel dashboard pages with KPI cards, charts, data bars, heatmaps and a key-insight box; consistent hotel colours; matching Power BI design.
Usability & user engagement (20%)	Excel: 4 shared drop-down filters, navigation buttons, "Showing" title, what-if and editable rules, PivotTable slicers and drill-down. Power BI: slicers, drill-through, tooltips, bookmarks.
Business impact & actionability (15%)	8 recommendations with evidence and estimated € impact; a risk flag on every booking; forecast for planning.
Data integration & management (5%)	5 source tables joined into a star schema; documented cleaning log; Power Query steps.
Performance & scalability (5%)	Measures (calculated columns only for the risk flag), star schema, incremental refresh plan, automatic risk flag on refresh, row-level security.

Table 18. Rubric alignment.

12. Limitations and Future Work

- **Lost revenue is an upper bound** – some cancelled rooms were resold. Actual loss is lower, but the relative comparisons still hold.
- **Only 26 months of data** – just two full seasonal cycles, so the forecast has limited history. More years would allow models such as Holt-Winters or Prophet.
- **No guest ID or room inventory** – occupancy % and RevPAR cannot be calculated because the number of rooms per hotel is not given.
- **Possible recording issues** in the parking and "Non Refund" fields (Section 7.2) – these were kept out of the risk flag.
- **The risk flag is a simple rule** – it is easy to explain but not optimised. It flags the riskiest bookings well (72% vs 19%) but misses some cancellations in the Low group.
- **Future work:** build a machine-learning cancellation model (predictive analytics) and compare it with the rule; add weather, events and competitor prices; publish the Power BI report with scheduled refresh; test a deposit policy as a controlled trial.

13. Conclusion

The project turned five raw tables into a working business analytics system. It shows that cancellations – not demand – are the hotels' biggest revenue problem, and that they are predictable: long lead times, group and OTA bookings, the domestic market and guests with no special requests carry most of the risk. The Excel workbook and Power BI report put these insights in front of managers, the risk flag shows which bookings to act on, and the forecast supports planning for the year ahead. Acting on the recommendations could recover millions of euros in booked value while making revenue more predictable.

References

- Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41–49. https://doi.org/10.1016/j.dib.2018.11.126
- Microsoft Learn. Power BI documentation – Key influencers visual and Analytics-pane forecasting. https://learn.microsoft.com/power-bi/ (accessed September 2026)
- Chapman, P. et al. (2000). CRISP-DM 1.0: Step-by-step data mining guide. SPSS Inc.

Appendix – Deliverables

File	Contents
Hotel_Booking_Analytics.xlsx	Main Excel dashboard (6 pages), forecast, risk flag, cleaning log, clean data
Excel_Dashboard_Guide.docx	How the Excel dashboard works, how to use and extend it, how to present it
PowerBI_Data (folder)	8 CSV tables for Power BI + colour theme JSON
PowerBI_Build_Guide.docx	Step-by-step Power BI instructions and all DAX measures
Hotel_Booking_Capstone_Presentation.pptx	Presentation slides
Hotel_Booking_Dashboard (web preview)	Interactive preview of the Power BI pages
Python scripts (clean.py, analyse.py, charts.py)	Reproducible cleaning, analysis, forecast, risk flag and report charts